

DUNE Interface Control Document: Slow Controls, Photon Detector System and Detector Protection System

Summary

This document defines the control, configuration and monitoring (CCM) and detector protection interfaces for DAPHNE, the Light Calibration Module and the PDS Power-over-Fiber (PoF) laser boxes. Each hardware system has its own physical-interface, service and protocol description. The interfaces cover PDS, Slow Controls (SC), the Detector Protection System (DPS) and DAQ where applicable. DPS provides protection infrastructure and detector protection.

Configuration and control use request/response; monitoring is published. The scope includes SC control and monitoring, DAQ use of DAPHNE and LCM, and DPS protection interfaces. DAQ data readout transport and optical timing transport are outside this scope.

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1 Responsibilities

PDS shall define equipment operating limits and protection requirements. DPS shall implement the agreed protection functions and publish their status. SC shall enforce operational permissions and publish detector availability and activity information. DAQ shall consume that information for run admission and data tagging.

Protection functions shall reference a versioned cause-and-effect matrix specifying the input source, protected equipment, action, response time, actuator feedback, failure behavior and reset conditions. Protective action shall remain effective independently of DAQ and the SC bridge. Section 5 contains the proposed PDS matrix.

PDS provides photon detector and light calibration hardware, firmware, native services for the warm electronics, detector specifications and detector operating restrictions. It shall provide monitoring information sufficient for SC to assess and confirm detector health and diagnose readout network connections, including SFP I²C diagnostics.

SC provides control and monitoring infrastructure, arbitrates access to detector controls and monitoring data, and assesses and publishes detector status. DPS provides protection infrastructure.

DAQ uses the detector, configures it, collects its data and provides calibration datasets agreed in advance with PDS. It shall check the status published by SC and take data only when SC advertises detector availability.

Ops arbitration (proposed; ownership to be agreed). SC completes the handshake with Operations (Ops) and advertises detector availability only after detector health and Ops authorization are confirmed.

2 Physical interface

2.1 DAPHNE

Each DAPHNE provides one dedicated 1 GbE SFP connection for CCM, as specified in the DAQ-PDS ICD [R1, R2]. Its board services share this link:

DAPHNE Ethernet interface $\longleftrightarrow$ switch $\longleftrightarrow$ physical server.

PDS supplies the fibres to its mini-rack patch panels; continuation and active switching are coordinated with DAQ and Technical Coordination [R1].

Detector	PDS channels	Channels/board	DAPHNE boards
FD-VD	1,344	32	42
FD-HD	6,000	40	150

These are the detector design allocations [R2], with one CCM connection per board. FD-VD includes membrane and cathode modules; both use this CCM interface.

The installation record shall identify board/switch/server ports, SFPs and patch routes. All DAPHNE CCM services share the board’s GbE bandwidth.

2.2 Light Calibration Module

Inventory under control and monitoring: 8 LCMs for FD-VD and 18 for FD-HD (planning quantities; PDS confirmation pending).

Each Light Calibration Module (LCM) has a dedicated 1 GbE SFP CCM connection [R2]: LCM Ethernet interface $\longleftrightarrow$ switch $\longleftrightarrow$ physical server. PDS shall specify the installed detector mapping, ports, media and patch routes.

2.3 PDS PoF laser boxes

Inventory under control and monitoring: 80 laser boxes for FD-VD (design allocation [R7]); none for FD-HD.

Power-over-Fiber (PoF) supplies the FD-VD cathode modules [R2]. A compact PLC system controls and monitors the laser boxes. The interface to the control network is Ethernet. PLC placement, laser-box I/O connections, Ethernet medium, rate, connectors and patch routes remain to be specified. PDS and DPS shall define the physical PLC/protection connection.

3 Protocols and services

3.1 DAPHNE

Location	Service chain
DAPHNE	AXI / OS $\rightarrow$ daphne-server $\rightarrow$ ZMQ.
SC bridge server	ZMQ client $\rightarrow$ OPC UA bridge.
SC server	OPC UA client $\rightarrow$ Ignition.
DAQ server	DAQ CCM client/opmon; separate Hermes configuration client.

The OPC UA bridge runs on the SC bridge server and includes DAQ configuration and control arbitration. Its ZMQ client receives **daphne-server** publications; the bridge exposes them to the SC server’s OPC UA client and to DAQ opmon. The SC server hosts Ignition. Hermes remains a separate DAPHNE service receiving configuration over UDP/IPBus. The installation record shall identify each service host and endpoint.

3.1.1 Configuration exchanges

DAQ CCM requests run-variable and local timing-endpoint configuration through the OPC UA bridge. The ZMQ client on the SC bridge server exchanges Protobuf messages

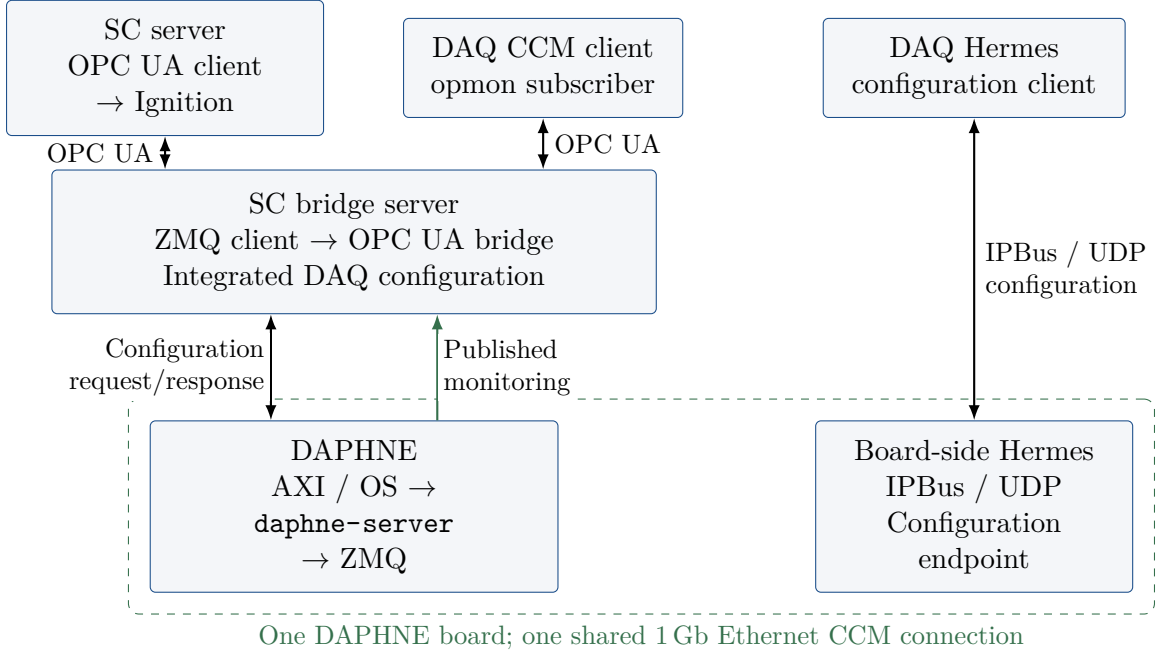


Figure 1: DAPHNE service locations and protocol paths.

over ZeroMQ and TCP using a DEALER client and the board’s ROUTER endpoint [R3, R4]. The reference port is TCP 40001 [R5]. Correlated responses shall distinguish acceptance, completion and failure.

Hermes retains its independent IPBus/UDP configuration endpoint, reference UDP 50001 [R6], outside the OPC UA bridge. Both services share the board connection and gateway lifecycle.

3.1.2 Published monitoring

Monitoring shall be published periodically or on change, without a client request for each update. Only configuration and control operations, including reset/recovery commands, use application request/response. Local hardware sampling is performed by the producing service.

The proposed DAPHNE publication transport is Protobuf over ZeroMQ PUB/SUB over TCP, using endpoints separate from configuration. The ZMQ client on the SC bridge server subscribes to `daphne-server` publications. The OPC UA bridge distributes them through OPC UA subscriptions to the SC server and DAQ opmon.

Consumer	Monitoring responsibility
DAQ opmon	Firmware counters, timing state and applied data-readout configuration, published by daphne-server through the OPC UA bridge.
SC / Ignition	Voltages, temperatures, fans, readout link diagnostics (including SFP I ² C monitoring), service status and software/firmware versions.
SC and DAQ	Timing-endpoint condition, reset/recovery progress and board readiness.

SC and opmon subscribe independently to their respective monitoring data through the OPC UA bridge. Publications shall carry source identity, time, sequence and validity. Subscribers shall detect stale or missing updates and obtain a current snapshot after reconnecting.

3.1.3 Timing-endpoint control, monitoring and recovery

DTS defines the endpoint controls and status. The endpoint HDL is in **daphne-firmware** [R8]; **daphne-os** provides the protocols for accessing its registers [R4]. The endpoint service reads and controls them locally; the OPC UA bridge shall expose this access to both DAQ and SC through GbE.

Timing configuration and monitoring use two paths. GbE carries local controls, including endpoint reset and address configuration, and publishes endpoint status and timing-SFP I²C status. The DTS optical fibre carries the remaining timing configuration and monitoring such as alignment status. Endpoint reset is required before the DTS configuration sequence.

GbE configuration and control use request/response; monitoring is published through OPC UA subscriptions.

A reset request shall identify its target and receive an execution response. The board shall publish reset assertion, release and the resulting endpoint state so that both clients can acknowledge the same recovery operation. Reset access shall remain available while timing is unavailable. Concurrent DAQ and SC commands shall be serialized by SC's control infrastructure.

SC shall determine and publish *board ready for data taking* from the agreed readiness conditions, including completed reset, restored clocks, endpoint readiness and valid synchronization [R8, R4]. Reset release alone is insufficient. The interface release shall define how SC receives readiness evidence carried by DTS. Missing or stale required publications shall withhold readiness. DAQ checks both board readiness and the detector availability published by SC before resuming data taking.

3.2 Light Calibration Module

Location	Calibration services
Light Calibration Module	PDS-provided LCM controller firmware and native calibration service.
SC physical server	Calibration OPC UA adapter and command arbitration.
SC infrastructure	Ignition client for equipment control and monitoring.
DAQ infrastructure	Client of the calibration interface for settings, sequences and status.

SC provides equipment power/service control and monitoring. The installation record shall identify each service host and endpoint.

Calibration requests and responses pass through the calibration OPC UA adapter to the LCM native service. That service shall publish equipment health, calibration status and readiness through OPC UA subscriptions to SC and DAQ. The control interface shall expose protective inhibit actions.

PDS shall specify the native control protocol, transport, ports, command acknowledgements, monitoring schema and compatible versions. These details remain open.

3.2.1 Timing-endpoint control, monitoring and recovery

LCMs use the same timing protocol and split configuration/monitoring paths as DAPHNE (Section 3.1.3). Both DAQ and SC access local endpoint configuration and GbE monitoring through the LCM service and OPC UA adapter. The same reset acknowledgement, command arbitration and access during timing outages apply. SC shall publish LCM readiness from fresh timing and recovery evidence; DAQ consumes it before calibration or data taking.

3.3 PDS PoF laser boxes

Location	PoF services
Compact PLC; placement to be specified	PoF control and monitoring logic.
Server location undecided	OPC UA server software for the compact PLC; implementation and driver to be selected.
SC infrastructure	Ignition OPC UA client for PoF control and monitoring.
DPS infrastructure	Detector protection; PLC/protection connection to be defined.

SC and PDS shall design the compact PLC system and identify and test its OPC UA server software. PDS provides PLC hardware and cables; SC integrates control and monitoring into Ignition.

The PLC-to-OPC-UA protocol, driver, transport, ports and server location are undecided. The selected implementation shall provide configuration/control request/response and publish monitoring to Ignition through OPC UA subscriptions. PoF has no DAQ interface. The interface shall expose DPS protection status and give protective inhibits priority over operating commands. PDS and DPS shall define the protection connection and acknowledgement behavior. The interface release shall specify the monitoring schema, compatible versions and service endpoints.

4 Interface release and verification

Each hardware system shall have an online interface specification and version record covering its firmware, services, clients, schemas, OPC UA model and endpoint/port allocation. Contracts shall define authentication, timeouts, queue limits, publication cadence and reconnect behavior. Configuration/control uses request/response; monitoring is published. For timing endpoints, record the DTS definitions and protocol revision and their mapping to local access and OPC UA.

DAPHNE: verify concurrent configuration and publication on the shared GbE link, HD/VD scales, subscriber reconnection and local endpoint controls against DTS definitions through OPC UA for DAQ and SC, including address configuration and acknowledged reset. Define the delivery of DTS readiness evidence to SC. Publication schemas and ports remain to be assigned.

Light Calibration Module: verify configuration and sequence requests, command arbitration, published status, compliance with DTS endpoint definitions, timing recovery and readiness, reconnection and inhibit acknowledgement.

PoF laser boxes: verify optical-power commands, arbitration, published operating/protection status, reconnection and inhibit acknowledgement.

The DAQ-PDS and DAQ-SC/DPS interface documents shall use these same contracts.

5 Proposed PDS interaction matrix

Hardware interlock. Automatically trips or inhibits affected equipment through hard-wired, local supply or autonomous DPS protection. It enforces an approved safe state independently of DAQ and the SC bridge.

Operational constraint. Governs permitted activities and state transitions through SC/Ops coordination. It may deny or hold a conflicting request; an operational conflict alone does not justify an equipment trip. Permitted overlaps are recorded with data-quality tags.

The following proposals retain the source matrix rule IDs [R9]. Each rule requires agreement by PDS, the affected subsystem and the responsible SC/DPS/DAQ owners.

Interaction / rule	Class	Proposed response
Purity monitor / Q-001	Data quality	Allow acquisition; SC publishes activity intervals and affected region for DAQ tagging.
Camera illumination / O-001	Operational exclusion	Deny the conflicting request; flag unexpected overlap and alert Ops.
Ionisation or alignment laser / O-002	Operational exclusion	Deny the conflicting request; require emission/shutter-off confirmation before PDS return. Flag unexpected overlap and alert Ops.
LCM calibration / Q-002	Calibration	Allow declared calibration; publish activity, configuration identity and affected scope for DAQ tagging.
PoF / C-001	Power dependency	Required for cathode PDS. Apply independent protection for its own faults.
Supply current or voltage fault / P-011	Equipment protection	Trip or inhibit the affected supply through the approved local/DPS path; verify its final state.
Equipment overtemperature / P-012	Equipment protection	Apply the approved local/DPS action to affected power, readout or calibration equipment.

The activity rules assign no PDS bias trip. Personnel laser safety remains independent. PoF has no DAQ interface.

Open for owner agreement: excluded PDS operating states; membrane and cathode scope; limits and response times; input validity and failure behavior; actuator mapping and feedback; settling, reset and recovery conditions.

References

- R1** DAQ–PDS ICD, 2621471: CCM physical connection.
- R2** DAQ–PDS proposed v8, 6c0b245: format, HD/VD inventory, configuration architecture, LCM CCM interface and ownership, cathode PoF supply.
- R3** daphnmodules, edf75a0: configuration schema and DEALER client.
- R4** daphne-os/daphne-server, e31bdfb: protocols for register access, ROUTER configuration endpoint and recovery prerequisites.
- R5** daphne-os/petalinux, e31bdfb: board services, endpoint initialization and lifecycle.
- R6** hermesmodules, 887d5fc: independent IPBus/UDP configuration service.
- R7** D. Leon, DUNE PoF design, FERMILAB-POSTER-24-0322-LBNF: FD-VD laser-box inventory.
- R8** daphne-firmware, 1f10fc0: DTS endpoint HDL, control and status definitions.
- R9** M. Arroyave, PDS interaction matrix, proposed v0.1: PDS rules adapted from the detector-wide interaction register of 3 September 2026 and the author’s ICD clarifications.